quantum nature. If you find the Planck constant in your equations, you know things are liable to get weird.

14 There's a subtlety here in this simplified explanation that has always bugged me. I'm telling you on the one hand that these regions have never communicated in the history of the cosmos, but I'm also telling you that the universe started with a singularity where, one might suppose, all the distances between things were zero. The reason this doesn't solve the problem is this. Take two points on opposite sides of the sky now. For the sake of argument, we'll say they were at zero distance at time zero. The problem is, at every time AFTER zero, those parts were not in contact—they couldn't have had any information exchange (like a light beam carrying information about temperature). What about, you say, zero itself? While we can label the first moment zero, it is literally zero time. Time began at the singularity. So there wasn't any time for the information exchange (because there wasn't any time) and every moment after that has the "too far apart to communicate" problem.

15 Today, we find antimatter in certain kinds of particle reactions, but mostly notice it because when an antimatter particle meets its matched regular matter particle, they annihilate, destroying each other and creating a burst of energy.

16 They are also, incidentally, giving us clues about the other end of time: recent breakthroughs have shown us evidence that the end of the universe might come in a totally unexpected way, and it may happen at any moment. But that's all covered later in the book; let's not get ahead of ourselves. We'll probably make it to Chapter 6.

17 I'm very sorry.

3. Big Crunch

1 Being the center of your own universe might sound appealing at first, until you factor in that the observational evidence for this is that everything is trying to get away from you as fast as possible.